

BRST Quantization and Representation Theory

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1 Introduction

The Becchi-Rouet-Stora-Tyutin (BRST) formalism provides a powerful cohomological framework for quantizing systems with symmetries. Its structure is deeply intertwined with the geometric process of symplectic reduction, a cornerstone in the study of physical models governed by symmetry principles. This project offers an exposition of the profound relationship between BRST theory and representation theory, with a particular focus on its application to vertex algebras.

We begin by establishing the algebraic foundations of group symmetries on Poisson manifolds [2], introducing Hamiltonian group actions and the resulting procedure of symplectic reduction. Following the seminal work of Kostant and Sternberg [5], we then frame BRST quantization as an algebraic analogue of this reduction.

Kostant and Sternberg introduced the concept of BRST quantization to the context of infinite dimensional Lie algebras, and related the obstruction of this construction to classical Lie algebra cohomology and the theory of representations of the Clifford algebra, there are various approaches to the understanding of these obstructions, which in modern terms are related to the theory of semi-infinite cohomology [6][3].

We present a key application of this theory in modern representation theory to the Virasoro algebra, following the ideas of Feign, reaching in a celebrated result from theoretical physics: the derivation of the critical spacetime dimension of 26 for bosonic string theory [8].

2 The BRST method

2.1 Hamiltonian Actions

Let (M, π) be a Poisson manifold and $\mathfrak{g}$ be a finite-dimensional Lie algebra.

We associate to every function in $C^\infty(M)$ a derivation and, equivalently, a vector field, the hamiltonian vector field.

$$\begin{aligned} C^\infty(M) &\rightarrow \mathfrak{X}(M) \\ f &\mapsto \{f, \cdot\} =: X_f \end{aligned}$$

Remember that an action of $\mathfrak{g}$ on M is a Lie morphism:

$$\begin{aligned} \mathfrak{g} &\rightarrow \mathfrak{X}(M) \\ u &\mapsto u_M \end{aligned}$$

We say that the action is **weakly hamiltonian** if it factors through a morphism $\hat{\mu} : \mathfrak{g} \rightarrow C^\infty(M)$, called the co-moment map, i.e, the following diagram commutes:

$$\begin{array}{ccc} & C^\infty(M) & \\ & \uparrow & \\ \mathfrak{g} & \longrightarrow & \mathfrak{X}(M) \end{array} \quad u_M = X_{\hat{\mu}(u)}$$

The moment map μ associated to the co-moment map $\widehat{\mu}$ is defined through:

$$\langle \mu(x), u \rangle := \widehat{\mu}(u)(x),$$

the fact that $\widehat{\mu}$ is linear implies that μ is well-defined, the natural inclusion:

$$\text{ev} : \mathfrak{g} \rightarrow C^\infty(\mathfrak{g}^*)$$

satisfies $\mu^* \circ \text{ev} = \widehat{\mu}$, and therefore, since $\mathfrak{g}^*$ is globally charted by $\text{ev}(\mathfrak{g})$, then:

$$\mu^* f \in C^\infty(M) \text{ for all } f \in \text{ev}(\mathfrak{g}) \Rightarrow \mu \text{ is smooth}$$

Proposition 1. [2, Exercise 3.11] μ is a Poisson map if, and only if, it is $\mathfrak{g}$ -equivariant.

Proof. The coadjoint action of $\mathfrak{g}$ on $\mathfrak{g}^*$ associates to $u \in \mathfrak{g}$, the derivation $u_{\mathfrak{g}^*} = \text{ad}_u^*$ satisfying:

$$\text{ad}_u^*(\text{ev}(v)) = \text{ev}([u, v])$$

.

$\mathfrak{g}$ -equivariance means that u_m is μ -related to $u_{\mathfrak{g}^*}$

$$u_M \stackrel{\mu}{\sim} u_{\mathfrak{g}^*}$$

i.e, that for every $f \in C^\infty(\mathfrak{g}^*)$ we have:

$$u_M(\mu^* f) = \mu^* u_{\mathfrak{g}^*}(f)$$

Again, since $\mathfrak{g}^*$ is a linear manifold it suffices to show the result for $f = \text{ev}(v)$, $v \in \mathfrak{g}$. On one hand:

$$\mu^*(u_{\mathfrak{g}^*}(\text{ev}(v))) = \mu^*(\{\text{ev}(u), \text{ev}(v)\}) = \mu^*(\text{ev}([u, v])) = \widehat{\mu}([u, v])$$

while on the other

$$u_M(\mu^* \text{ev}(v)) = X_{\widehat{\mu}(u)}(\widehat{\mu}(v)) = \{\widehat{\mu}(u), \widehat{\mu}(v)\}$$

Which are the same, if and only if, μ is a Poisson map. □

In the case where μ is a Poisson map we say that the action is **hamiltonian**.

2.2 Symplectic Reduction

Assume that the action of $\mathfrak{g}$ on M is particularly nice, it is the differentiation of a free and proper action of a Lie group G integrating $\mathfrak{g}$, assume also that M is symplectic.

Given an element $\xi \in \mathfrak{g}^*$ we denote by G_ξ the isotropy group $\{g \in G : g\xi = \xi\}$

Theorem 2. [2, p.33, 34] Under the assumptions above, M/G is a Poisson manifold with symplectic foliation by connected components of

$$\mu^{-1}(\xi)/G_\xi$$

Proof. Since the action is assumed to be free and proper, then the quotient map

$$p : M \rightarrow M/G =: N$$

is a submersion. The induced morphism on functions is injective and has as image the ring of invariants.

$$C^\infty(M/G) \cong C^\infty(M)^G = \{f \in C^\infty(M) : f(g \cdot x) = f(x) \quad \forall g \in G, x \in M\}$$

We'll show that this gives rise to a Poisson structure on M/G as a complete and full symplectic realization. The symplectic leaves are known in such a case: they are the integration of the distribution ν^ω pushed to M/G by p , ν is the distribution tangent to the orbits of M .

Lemma 3. *If G is connected and acts smoothly on a symplectic manifold with infinitesimal action being hamiltonian, then G acts by symplectomorphisms.*

Idea of proof. We want to show that the map $L_g : M \rightarrow M$ is a symplectomorphism, without loss of generality one may take $g = \exp(tu)$. Then the action of $\exp(tu)$ is the flow of the infinitesimal generator u_M , which is a hamiltonian field and therefore preserves the symplectic form. $\square$

Therefore $C^\infty(M)^G$ is a Poisson subalgebra of $C^\infty(M)$:

$$L_g^*\{f, g\} = \{L_g^*f, L_g^*g\} = \{f, g\} \quad \text{for all } f, g \in C^\infty(M)^G$$

Lemma 4. *The map $p : M \rightarrow N$ is a complete full symplectic realization of N and satisfies $(\ker dp)^\omega = \ker d\mu$*

Idea of proof. The fact that it is a full symplectic realization follows from the previous lemma (and from p being a surjective submersion), it is complete if for any hamiltonian field on $X_f \in \mathfrak{X}(N)$ with flow determined for infinite time, then $X_{p^*f} \in \mathfrak{X}(M)$ also has flow determined in infinite time, this is [2, Exercise 4.14], the idea is that if the flow of p^*f starting at $x \in M$ stopped at some point $y \in M$, we may use the group action to transport the flow of X_f to some point $z \in M$ in the same orbit as y , and use the group action to transport the flow at z to y .

Now for the desired property, consider that $\ker(dp)$ and $\ker(d\mu)$ have complementary dimensions so it suffices to show that $\ker(d\mu) \subseteq \ker(dp)^\omega$, fixed a point $x \in M$, $X \in \ker(d\mu)|_x$ and $Y \in \ker(dp)|_x$, then, by the fact that Y is tangent to the orbit at x , there exists some element $u \in \mathfrak{g}$ such that $Y|_x = u_M|_x$, the fact that the action is hamiltonian shows that:

$$\omega(Y, X) = i_Y\omega(X) = d\langle u, \mu(x) \rangle(X) = 0$$

Showing the desired result. $\square$

$\square$

2.3 BRST Cohomology

We want to quantize the process of symplectic reduction, to do so, we first give an alternate description of symplectic reduction using tools from homological algebra.

The setup is as follows: We have a hamiltonian free and proper action of a Lie group G on a symplectic manifold (M, ω) with moment map μ .

Proposition 5. *Under the restriction map $C^\infty(M) \rightarrow C^\infty(\mu^{-1}(0))$, we have that*

$$C^\infty(\mu^{-1}(0)) \cong C^\infty(M)/I$$

where $I = \langle \widehat{\mu}(\mathfrak{g}) \rangle$.

Idea of proof. Since every smooth function on a submanifold can be extended, the map $C^\infty(M) \rightarrow C^\infty(\mu^{-1}(0))$ is surjective, its kernel is exactly:

$$I = \{f \in C^\infty(M) : f|_{\mu^{-1}(0)} = 0\}$$

Of course $\widehat{\mu}(\mathfrak{g}) \subseteq I$ since $\mu^*u(p) = u(0) = 0$ by linearity. Now, the fact that the ideal I is generated by such functions follows from the fact that the map $\mu : M \rightarrow \mathfrak{g}^*$ is a submersion and we can apply a version of Taylor's theorem called Hadamard's lemma to show that the result holds locally and extend to the whole of $\mu^{-1}(0)$ by partitions of unity.

Therefore we have a construction of $\mu^{-1}(0)$ in purely algebraic terms. By theorem 2 we have that $C^\infty(\mu^{-1}(0)/G)$ can be described by the invariants of $C^\infty(\mu^{-1}(0))$, which can be further described through Lie algebra cohomology, so that:

$$C^\infty(\mu^{-1}(0)/G) = C^\infty(\mu^{-1}(0))^G = H^0(\mathfrak{g}, \mu^{-1}(0))$$

The construction of $C^\infty(\mu^{-1}(0))$ can be refined further using a tool from homological algebra, the so called Koszul Complex:

Lemma 6. *(Koszul Complex [7]) Let R be a commutative algebra, M an R -algebra and $\delta : R^n \rightarrow M$ module morphism, then we have the associated Koszul complex:*

$$\left(\bigwedge^\bullet R^n \otimes M, \delta \right)$$

Where δ is defined as:

$$\begin{aligned} \delta : \bigwedge^k R^n \otimes R &\rightarrow \bigwedge^{k-1} R^n \otimes R \\ (e_{i_1} \wedge \cdots \wedge e_{i_k} \otimes f) &\mapsto \sum_{j=1}^k (-1)^{j+1} e_{i_1} \wedge \cdots \wedge \widehat{e_{i_j}} \wedge \cdots \wedge e_{i_k} \otimes \delta(e_{i_j})f \end{aligned}$$

Under regularity conditions for M [1, Lemma 7.5], the complex satisfies:

$$H^i \left(\bigwedge^\bullet R^n \otimes M \right) = \begin{cases} M / \langle \delta(M) \rangle, & \text{if } i = 0 \\ 0, & \text{otherwise.} \end{cases}$$

Using Lie algebra cohomology and the Koszul complex, we have a double complex

$$C^{p,q} = \bigwedge^p \mathfrak{g}^* \otimes \bigwedge^q \mathfrak{g} \otimes C^\infty(M)$$

A simple calculation shows that the Chevalley-Eilenberg differential d commutes with the Koszul differential δ in every degree, and the associated spectral sequence stabilizes at the second page under the regularity conditions for the Koszul Complex which are satisfied in our situation.

Page 0:

$$\begin{array}{ccccccc}
 C^{0,q} & \longrightarrow & C^{0,q-1} & \longrightarrow & \dots & \longrightarrow & C^{0,0} \\
 \\
 C^{1,q} & \longrightarrow & C^{1,q-1} & \longrightarrow & \dots & \longrightarrow & C^{1,0} \\
 \\
 \vdots & & \vdots & & \ddots & & \vdots \\
 \\
 C^{p,q} & \longrightarrow & C^{p,q-1} & \longrightarrow & \dots & \longrightarrow & C^{p,0}
 \end{array}$$

Page 1:

$$\begin{array}{ccccccc}
 0 & 0 & 0 & & C^\infty(\mu^{-1}(0)) \\
 \downarrow & \downarrow & \downarrow & & \downarrow \\
 0 & 0 & 0 & & \mathfrak{g}^* \otimes C^\infty(\mu^{-1}(0)) \\
 \downarrow & \downarrow & \downarrow & & \downarrow \\
 \vdots & \vdots & \vdots & & \vdots \\
 \downarrow & \downarrow & \downarrow & & \downarrow \\
 0 & 0 & 0 & & \bigwedge^p \mathfrak{g}^* \otimes C^\infty(\mu^{-1}(0))
 \end{array}$$

Page 2:

$$\begin{array}{ccccccc}
 0 & 0 & 0 & & C^\infty(\mu^{-1}(0)/G) & & 0 \\
 \searrow & \searrow & \searrow & & \searrow & & \searrow \\
 0 & 0 & 0 & & H^1(\mathfrak{g}, C^\infty(\mu^{-1}(0))) & & 0 \\
 \searrow & \searrow & \searrow & & \searrow & & \searrow \\
 \vdots & \vdots & \vdots & & \vdots & & 0 \\
 \searrow & \searrow & \searrow & & \searrow & & \searrow \\
 0 & 0 & 0 & & H^p(\mathfrak{g}, C^\infty(\mu^{-1}(0))) & & 0
 \end{array}$$

The fact that this spectral sequence stabilizes shows that the double complex:

$$C_{BRST}^n = \bigoplus_{p-q=n} \bigwedge^p \mathfrak{g}^* \otimes \bigwedge^q \mathfrak{g} \otimes C^\infty(M)$$

With differential $D = d + (-1)^p \delta$ satisfies:

$$H_{BRST}^p(C^\bullet) = H^p(\mathfrak{g}, H^0(C^{p,\bullet}, \delta))$$

3 Quantization

Having established the classical framework of Hamiltonian systems using the language of Poisson geometry, we now turn to the pivotal process of quantization. At its core, quantization is a procedure for constructing a quantum theory from a classical one. In the algebraic setting, this corresponds to a non-commutative algebra of quantum operators. This section formalizes this transition, beginning with the structure of Super Poisson algebras, which provide the necessary language to carry the classical symmetries discussed previously into the quantum domain.

3.1 Super Poisson Algebras

We want to view this construction in a more intrinsically algebraic way, for such, we talk a bit about Poisson super-algebras.

A **super-algebra** is an algebra $(\mathcal{A}, \cdot)$ with a $\mathbb{Z}/2\mathbb{Z}$ grading:

$$\mathcal{A} = \underbrace{\mathcal{A}_0}_{\text{even}} \oplus \underbrace{\mathcal{A}_1}_{\text{odd}},$$

the compatibility with the product is given through $A_i \cdot A_j \subseteq A_{i+j}$.

We denote the parity of an element $a \in \mathcal{A}$ by $|a|$.

If $\mathcal{A}$ is associative then the super-commutator:

$$[a, b] = ab - (-1)^{|a||b|}ba$$

satisfies the super analogues of skew symmetry and Jacobi:

$$[a, b] = (-1)^{|a||b|}[b, a] \quad [a, [b, c]] = [[a, b], c] + (-1)^{|a||b|}[b, [a, c]]$$

The algebra is said to be **super-commutative** if $[a, b] = 0$ for every $a, b \in \mathcal{A}$.

Examples:

- Any algebra is a super-algebra with every element being even.
- The Grassmann algebra $\bigwedge V$ is a super-commutative super-algebra, with k -vectors having the same parity as k .

An algebra with product satisfying the super-analogues of skew-symmetry and Jacobi is a **Lie super-algebra**.

A **super-Poisson** algebra is a super-commutative algebra with an additional Lie super-algebra structure compatible by the super analogue of Leibniz:

$$\{a, b \cdot c\} = \{a, b\} \cdot c + (-1)^{|a||b|} b \cdot \{a, c\}$$

Super Poisson algebras generalize the concept of usual Poisson algebras (a Poisson algebra being simply a super Poisson algebra with only even elements).

Remark: When doing calculation in a super-commutative algebra or remembering the identities, it reduces to adding the term $(-1)^{|x||y|}$ whenever you swap x and y .

Examples:

- Given a Lie algebra $\mathfrak{g}$, the universal enveloping algebra $U(\mathfrak{g})$ is an associative algebra and, therefore, a super-Poisson algebra with the Poisson bracket given by the commutator.
By the PBW theorem, this algebra is a quantization of the classical algebraic Poisson manifold $\text{Spec } \mathbb{C}[\mathfrak{g}^*]$ corresponding to the usual Poisson manifold $\mathfrak{g}^*$.
- For a vector space V with a symmetric bilinear form $(\cdot, \cdot)$, then we can construct the Clifford algebra $\mathcal{Cl}(V, (\cdot, \cdot))$, it is an associative algebra which corresponds to a quantization of the Grassman-algebra $\bigwedge V$.

In our case we have a complex that can be given a natural super commutative structure:

$$C_D^\bullet = \bigwedge (\mathfrak{g} \oplus \mathfrak{g}^*) \otimes R$$

We have the usual Poisson structure on R , we are missing the ingredient of a super-Poisson structure on $\bigwedge (\mathfrak{g} \oplus \mathfrak{g}^*)$. This structure is given through the bilinear form induced on $\mathfrak{g} \oplus \mathfrak{g}^*$:

$$(f + x, g + y) = f(y) + g(x)$$

This induces a map $l : \mathfrak{g} \oplus \mathfrak{g}^* \rightarrow (\mathfrak{g} \oplus \mathfrak{g}^*)^*$, we define the super Poisson bracket through the interior product ι :

$$\{x, \alpha\} = \iota_{l(x)} \alpha$$

And extending this to be a derivation.

Now we have a super-Poisson algebra $\mathcal{A}$ for $C_D^\bullet$. we want to find an element $\Theta \in \mathcal{A}$ such that:

$$\{\Theta, x\} = Dx$$

[5] shows that for a given basis e_i of $\mathfrak{g}$ with dual basis e_i^* and structure constants c_{ij}^k :

$$[e_i, e_j] = \sum_k c_{ij}^k e_k$$

Then

$$\Omega = \sum_{i,j,k} c_{ij}^k e_i \wedge e_j^* \wedge e_k^* \in \bigwedge (\mathfrak{g} \oplus \mathfrak{g}^*)$$

And

$$\tilde{\delta} = \sum_{i=1}^n y_i \otimes \hat{\mu}(x_i)$$

correspond to d and δ , respectively, and:

$$\Theta = \Omega \otimes 1 + \tilde{\delta}$$

represents the differential D .

Theorem 7.

$$\{\Theta, \bullet\} = D$$

3.2 The Quantum Analog

The quantization of Θ is then given part by part.

The quantization of $\bigwedge^\bullet(\mathfrak{g} \oplus \mathfrak{g}^*)$ is the Clifford algebra $\mathcal{Cl}(\mathfrak{g} \oplus \mathfrak{g}^*)$, given a quantization of our phase space $\mathcal{H}$, with a map:

$$\tau : C^\infty(M) \rightarrow \text{End}(\mathcal{H})$$

Then the quantization of $\Omega \otimes 1$ is some element $Q \in \mathcal{Cl}(\mathfrak{g} \oplus \mathfrak{g}^*)$, and the quantization of δ is given by $\tau \in \text{End}(\mathcal{H})$, so that:

$$\mathbf{Q} = -\frac{1}{2}Q \otimes 1 + \tau$$

Here

$$Q = \sum_{i,j,k} c_{ij}^k (e_k e_i^* e_j^*)$$

and

$$\tau = \sum_{i=1}^n e_i^* \otimes \tau(\widehat{\mu}(e_i))$$

Theorem 8. $Q^2 = 0$

Denoting by ρ the composition:

$$\begin{aligned} \rho : \mathfrak{g} &\rightarrow C^\infty(M) \rightarrow \text{End}(\mathcal{H}) \\ u &\rightarrow \widehat{\mu}(u) \rightarrow \tau(\widehat{\mu}(u)) \end{aligned}$$

Observe that, since $\tau : C^\infty(M) \rightarrow \text{End}(\mathcal{H})$ is a quantization and $\widehat{\mu}$ is a Lie map, then ρ is a Lie map.

$$\mathbf{Q}^2 = \underbrace{\tau^2}_{(I)} - \frac{1}{2} \underbrace{(\tau Q + Q \tau)}_{(II)} + \frac{1}{4} \underbrace{Q^2}_{(III)} \otimes 1;$$

We shall show in what follows, by direct computations in the Clifford algebra, that:

$$2(II) = (I) \quad (III) = 0$$

(I) :

$$\begin{aligned} \tau^2 &= \sum_{i,j=1}^n e_i^* e_j^* \otimes \rho(e_i) \rho(e_j) && \text{Split sum in two} \\ &= \frac{1}{2} \sum_{i,j=1}^n e_i^* e_j^* \otimes \rho(e_i) \rho(e_j) + e_j^* e_i^* \otimes \rho(e_j) \rho(e_i) && \text{Swap } e_j^* e_i^* \text{ with } -e_i^* e_j^* \\ &= \frac{1}{2} \sum_{i,j=1}^n e_i^* e_j^* \otimes (\rho(e_i) \rho(e_j) - \rho(e_j) \rho(e_i)) && \rho \text{ is a representation} \\ &= \frac{1}{2} \sum_{i,j=1}^n e_i^* e_j^* \otimes \rho([e_i, e_j]) \end{aligned}$$

(II) :

$$\begin{aligned}
\tau Q + Q\tau &= \sum_{i,j,k,l=1}^n (c_{jk}^l e_i^* e_j^* e_k^* e_l + c_{jk}^l e_j^* e_k^* e_l e_i^*) \otimes \rho(e_i) & e_i^* e_j^* e_k^* &= e_j^* e_k^* e_i^* \\
&= \sum_{i,j,k,l=1}^n (c_{jk}^l e_j^* e_k^* (e_i^* e_l - e_l e_i^*)) \otimes \rho(e_i) & e_i^* e_l + e_l e_i^* &= \delta_{il} \\
&= \sum_{j,k,l=1}^n e_j^* e_k^* \otimes \rho(c_{jk}^l e_l) & [e_j, e_k] &= \sum_{l=1}^n c_{jk}^l e_l \\
&= \sum_{j,k=1}^n e_j^* e_k \otimes \rho([e_j, e_k])
\end{aligned}$$

Proving our first assertion, as for the second one, we need a lemma:

Lemma 9. *Define*

$$\begin{aligned}
\theta : \mathfrak{g} &\rightarrow \mathcal{Cl}(\mathfrak{g} \oplus \mathfrak{g}^*) \\
x &\mapsto \sum_{i=1}^n [x, e_i] e_i^*
\end{aligned}$$

Then, for all $x, y \in \mathfrak{g}$ with $z = [x, y]$ we have:

- (a) $[\theta(x), y] = z$
- (b) $[\theta(x), \theta(y)] = \theta(z)$
- (c) $[Q, x] = -2\theta(x) + 2 \operatorname{Tr}(ad(x)).$

Proof. These are direct calculations on the Clifford algebra:

- (a) Without loss of generality, let $x = e_i$ and $y = e_j$

$$\begin{aligned}
\theta(x)y &= \sum_{k=1}^n [e_i, e_k] e_k^* e_j & e_k^* e_j + e_j e_k^* &= \delta_{jk} \\
&= \sum_{k=1}^n [e_i, e_k] \delta_{jk} - \sum_{k=1}^n [e_i, e_k] e_j e_k^* & [e_i, e_k] e_j &= -e_j [e_i, e_k] \\
&= [e_i, e_j] - \sum_{k=1}^n e_j [e_i, e_k] e_k^* & [e_i, e_j] &= z \\
&= z - y\theta(x)
\end{aligned}$$

(b) Without loss of generality, let $x = e_i$ and $y = e_j$

$$\begin{aligned}
\theta(x)\theta(y) &= \sum_{k,l=1}^n [e_i, e_k] e_k^* [e_j, e_l] e_l^* & [e_j, e_l] &= \sum_{m=1}^n e_{jl}^m e_m \\
&= \sum_{k,l,m=1}^n [e_i, e_k] e_k^* e_m e_l^* & & e_k^* e_m + e_m e_k^* \delta_{km} \\
&= \sum_{k,l,m=1}^n c_{jl}^m \delta_{km} [e_i, e_k] e_l^* - \sum_{k,l,m=1}^n c_{jl}^m [e_i, e_k] e_m e_k^* e_l^* \\
&= \sum_{l,m=1}^n [e_i, \sum_{m=1}^n c_{jl}^m e_m] e_l^* - \sum_{k,l,m=1}^n c_{jl}^m e_m [e_i, e_k] e_l^* e_k^* & c_{jl}^m e_m &= [e_j, e_l] \\
&= \sum_{l=1}^n [e_i, [e_j, e_l]] e_l^* - \sum_{k,l,m=1}^n c_{ik}^m [e_j, e_l] e_m e_l^* e_k^* & e_m e_l^* + e_l^* e_m &= \delta_{ml} \\
&= \sum_{k=1}^n [e_i, [e_j, e_k]] e_k^* - \sum_{k,l,m=1}^n c_{ik}^m \delta_{ml} [e_j, e_l] e_k^* + \theta(y)\theta(x) & c_{ik}^m \delta_{ml} e_l &= [e_i, e_k] \\
&= \sum_{k=1}^n ([e_i, [e_j, e_k]] - [e_j, [e_i, e_k]]) e_k^* + \theta(y)\theta(x) & & \text{Jacobi} \\
&= \sum_{k=1}^n [[e_i, e_j], e_k] e_k^* + \theta(y)\theta(x) \\
&= \theta([x, y]) + \theta(y)\theta(x)
\end{aligned}$$

(c) Without loss of generality, let $x = e_l$

$$\begin{aligned}
Qx &= \sum_{i,j,k=1}^n c_{ij}^k e_k e_i^* e_j^* e_l \\
&= - \sum_{i,j=1}^n [e_i, e_j] e_j^* e_i^* e_l \\
&= - \sum_{i=1}^n \theta(e_i) e_i^* e_l \\
&= -\theta(e_l) + \sum_{i=1}^n \theta(e_i) e_l e_i^* \\
&= -\theta(e_l) - xQ
\end{aligned}$$

$$\begin{aligned}
\tilde{Q}x &= \sum_{i,j,k=1}^n c_{ij}^k e_i^* e_j^* e_k e_l & e_k e_l &= -e_l e_k \\
&= - \sum_{i,j,k=1}^n c_{ij}^k e_i^* e_j^* e_l e_k & e_j^* e_l &= \delta_{jl} - e_l e_j^* \\
&= - \sum_{i,j,k=1}^n c_{ij}^k \delta_{jl} e_i^* e_k + \sum_{i,j,k=1}^n c_{ij}^k e_i^* e_l e_j^* e_k & e_i^* e_l &= \delta_{il} - e_l e_i^* \\
&= - \sum_{i,k=1}^n c_{il}^k e_i^* e_k + \sum_{j,k=1}^n c_{lj}^k e_j^* e_k - x\tilde{Q} & e_i^* e_k &= \delta_{ik} - e_k e_i^* \\
&= - \sum_{i,j=1}^n c_{il}^k \delta_{ik} + \sum_{i,k=1}^n c_{il}^k e_k e_i^* + \sum_{j,k=1}^n c_{lj}^k \delta_{kj} - \sum_{j,k=1}^n c_{lj}^k e_k e_j^* - x\tilde{Q} \\
&= -2 \sum_{i=1}^n [e_l, e_i] e_i^* + 2 \sum_{i=1}^n c_{li}^i - x\tilde{Q} \\
&= -2\theta(x) + 2\text{Tr}(\text{ad}(x)) - x\tilde{Q} \quad \square
\end{aligned}$$

Using this lemma, consider, for all $x, y \in \mathfrak{g}$, defining $z = [x, y]$:

$$\begin{aligned}
[x, [y, [Q, Q]]] &= [x, 2[[y, Q], Q]] & (c) \\
&= -2[x, [2\theta(y), Q]] & \text{Jacobi} \\
&= -4([x, \theta(y)], Q) + [\theta(y), [x, Q]] & (a) \text{ \& } (c) \\
&= 4([z, Q] - 2[\theta(y), \theta(x)]) & (b) \text{ \& } (c) \\
&= -8\theta(z) + 8\theta(z) + 8\text{Tr}(\text{ad}(z)) \\
&= 0
\end{aligned}$$

Therefore $Q^2 = \frac{1}{2}[Q, Q]$ satisfies $[x, [y, Q^2]] = 0 \Rightarrow [y, Q^2] \in \bigwedge \mathfrak{g}$ but then $Q^2 \in \bigwedge \mathfrak{g}$ and, finally, $Q^2 = 0$ as $\deg Q^2 = 2$ and $\deg \bigwedge \mathfrak{g} < 0$ when considering the grading $\mathfrak{g}$ at -1 and $\mathfrak{g}^*$ at 1 . $\square$

4 Bosonic Strings

4.1 BRST in the infinite-dimensional setting

There are various problems with trying to directly expand the BRST construction to infinite dimensional Lie algebras and their modules, the approach that is generally done in this setting is that of semi-infinite cohomology. In this section we provide an exposition of the process of normalization and the semi-infinite in an elementary way (this follows [4]).

The problems:

- $\mathfrak{g}^*$ does not have a countable basis.
This is solved through the introduction of a topology on $\mathfrak{g}$, algebraically, this is done through a $\mathbb{Z}$ -grading on $\mathfrak{g}$ and restricting $\mathfrak{g}^*$ to continuous morphisms with respect to this topology (the restricted dual is usually denoted by $\mathfrak{g}^{\otimes}$).
- The element Q is not well defined as an infinite sum, not even in a formal way.
This is solved through normal ordering. We represent the element Q not as acting on $\mathcal{C}\ell(\mathfrak{g} \oplus \mathfrak{g}^{\otimes}) \cong \bigwedge(\mathfrak{g} \otimes \mathfrak{g}^{\otimes})$ but acting on a natural ‘limit’ known as the space of semi-infinite forms or, sometimes called the ‘Dirac-sea’.
- There is an homological obstruction for the existence of the representation θ from Lemma 9 (and, therefore, for the vanishing of Q^2).
There is no easy solution to this problem, so we are forced to reduce the study to $\mathbb{Z}$ -graded Lie algebras for which this homological obstruction does not exist.

So, fix a $\mathbb{Z}$ -graded Lie algebra $\mathfrak{g} = \bigoplus \mathfrak{g}_n$, the space of semi-infinite forms is the induced space:

$$\bigwedge^{\infty/2} \mathfrak{g} = \mathcal{C}\ell(\mathfrak{g} \oplus \mathfrak{g}^{\otimes}) \otimes_S |0\rangle$$

where S is the subalgebra $S = \mathcal{C}\ell(\mathfrak{g}_{\leq 0} \oplus \mathfrak{g}_{> 0}^{\otimes})$ acting trivially on $|0\rangle$, define, for $\gamma \in \mathfrak{g}_0^{\otimes}$ and $x \in \mathfrak{g}$ the linear map:

$$\theta_\gamma(x) = \gamma(x) + \sum_i :e_i^*[e_i, x]:$$

The claim is that

$$[\theta_\gamma(x), \theta_\gamma(y)] - \theta_\gamma([x, y])$$

is a cocycle $\omega + d\gamma$ (called the critical cocycle) and that, if $d\gamma = -\omega$ then it is a true representation of $\mathfrak{g}$, sometimes called a semi-infinite structure on $\mathfrak{g}$.

If $\mathfrak{g}$ has semi-infinite structure θ with cocycle δ , and V is another $\mathfrak{g}$ -module (from category $\mathcal{O}$), then:

$$Q = \text{id}_V \otimes \gamma + \sum_i e_i \otimes e_i^* - \frac{1}{2} \sum_{i,j,k} c_{ij}^k :e_k e_i^* e_j^*: \in \text{End} \left(V \otimes \bigwedge^{\infty/2} \mathfrak{g} \right)$$

satisfies $Q^2 = 0$.

4.2 Conformal Theory and the Virasoro Algebra

Defining the vector fields:

$$L_n = z^{n+1} \frac{d}{dz} \quad n \in \mathbb{Z}$$

defined on $\mathbb{C} \setminus \{0\}$, then the Witt-algebra $W = \text{Span}_{\mathbb{C}} \{L_n : n \in \mathbb{Z}\}$ is the algebra of infinitesimal conformal transformations in $\mathbb{R}^2$ and satisfies the bracket relations:

$$[L_m, L_n] = (m - n)L_{m+n}$$

so W is $\mathbb{Z}$ -graded. It is well-known that this algebra has a single central extension, i.e, that there exists a single element $\omega \in \wedge^2 \mathfrak{g}^{\otimes}$ such that $H^2(W, \mathbb{C}) = \omega$. This central extension is called the Virasoro algebra and, by convention (related to the ζ -function and vertex algebras) we write the bracket on that algebra as:

$$[L_m, L_n] = (m - n)L_{m+n} + \delta_{m,-n} \frac{m^3 - m}{12} C$$

4.3 Quantization and 26

Applying the above procedure to the virasoro algebra, one can show that there is only one semi-infinite structure on Vir , and the associated element $\gamma \in \mathfrak{g}_0^{\otimes}$ is given by:

$$\gamma(C) = 26 \quad \gamma(L_0) = -2.$$

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